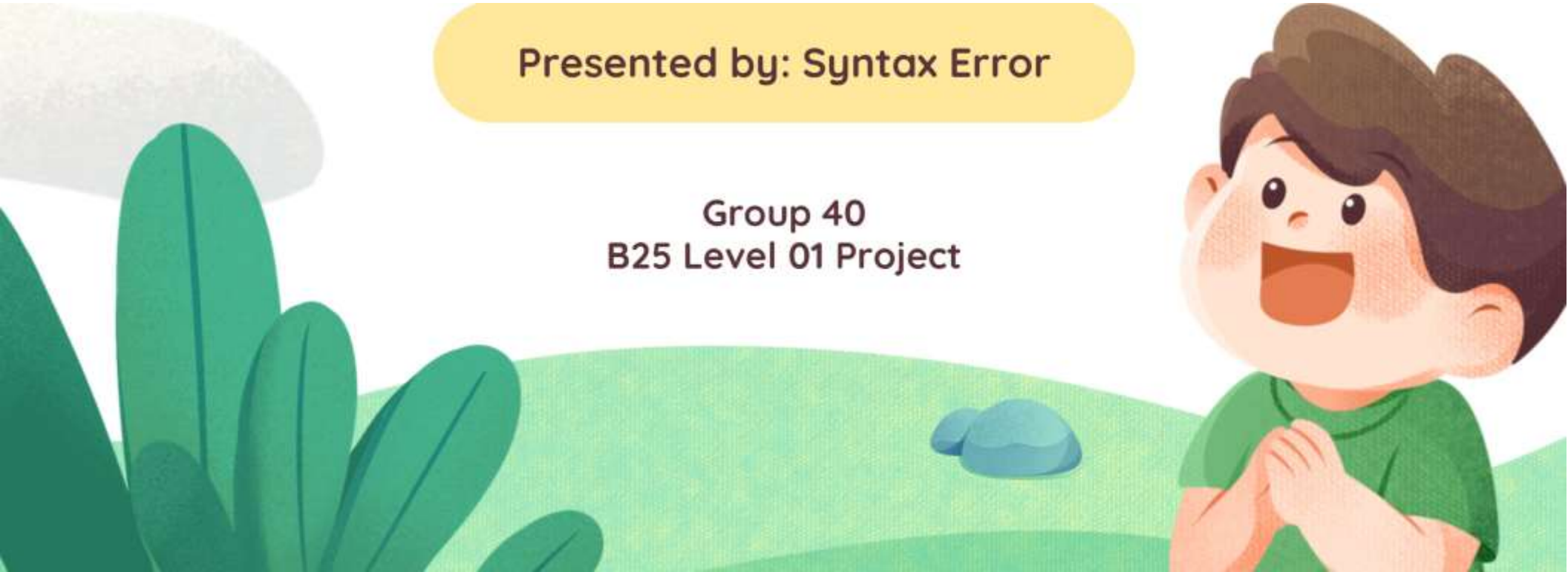
AQUANEXUS

Presented by: Syntax Error

Group 40
B25 Level 01 Project



TEAN MEMBERS

INDEX	NAME	DEGREE
254237M	Welhena WBR(Group Leader	I
254115M	Koralage KCCS	
255045F	Jayathilaka HPGKV	IT&M
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INTRODUCTION

AqUaCUItUre is a rapidly growing industry, but fish farming SUCCeSS depends heavily on water quality. Parameters such as pH, turbidity, dissolved solids, and temperature directly influence fish health and productivity. Traditional manual monitoring is inefficient, prone to human error, and unable to respond to sudden changes. AquaNexus proposes a smart, automated monitoring system that integrates sensors, microcontrollers, and intelligent algorithms to ensure real-time water quality analysis, early detection of unsafe conditions, and

M& OBJECTIVES

- Aim: To design and implement a low-cost, intelligent aqUaCUItUre monitoring system capable of real-time water quality analysis and automated alerts.

Objectives:

Monitor critical water parameters (pH, turbidity, TDS, temperature, water level, flow rate).
Detect unsafe conditions using threshold-based and Aldriven logic.
Provide real-time alerts via buzzer, LEDs, and LCD display. Simulate and validate system performance in a lab-scale prototype.
Ensure scalability and cost efficiency for small-scale fish farmers.

PROBLEM &

PROPOSED SOLUTION

Problem:

Rapid water quality fluctuations cause fish mortality.

Manual monitoring is time-consuming and inaccurate

Advanced commercial systems are expensive and inaccessible to small farmers.

SOLUTION

Develop a microcontroller-based system (Arduino Uno/Mega).

Integrate multiple sensors for comprehensive monitoring.

Use AI-inspired decision logic for smarter alerts (e.g., combining pH + turbidity readings).

Provide real-time feedback via LCD, buzzer, and LED indicators.

Build a lab prototype to validate performance before scaling.

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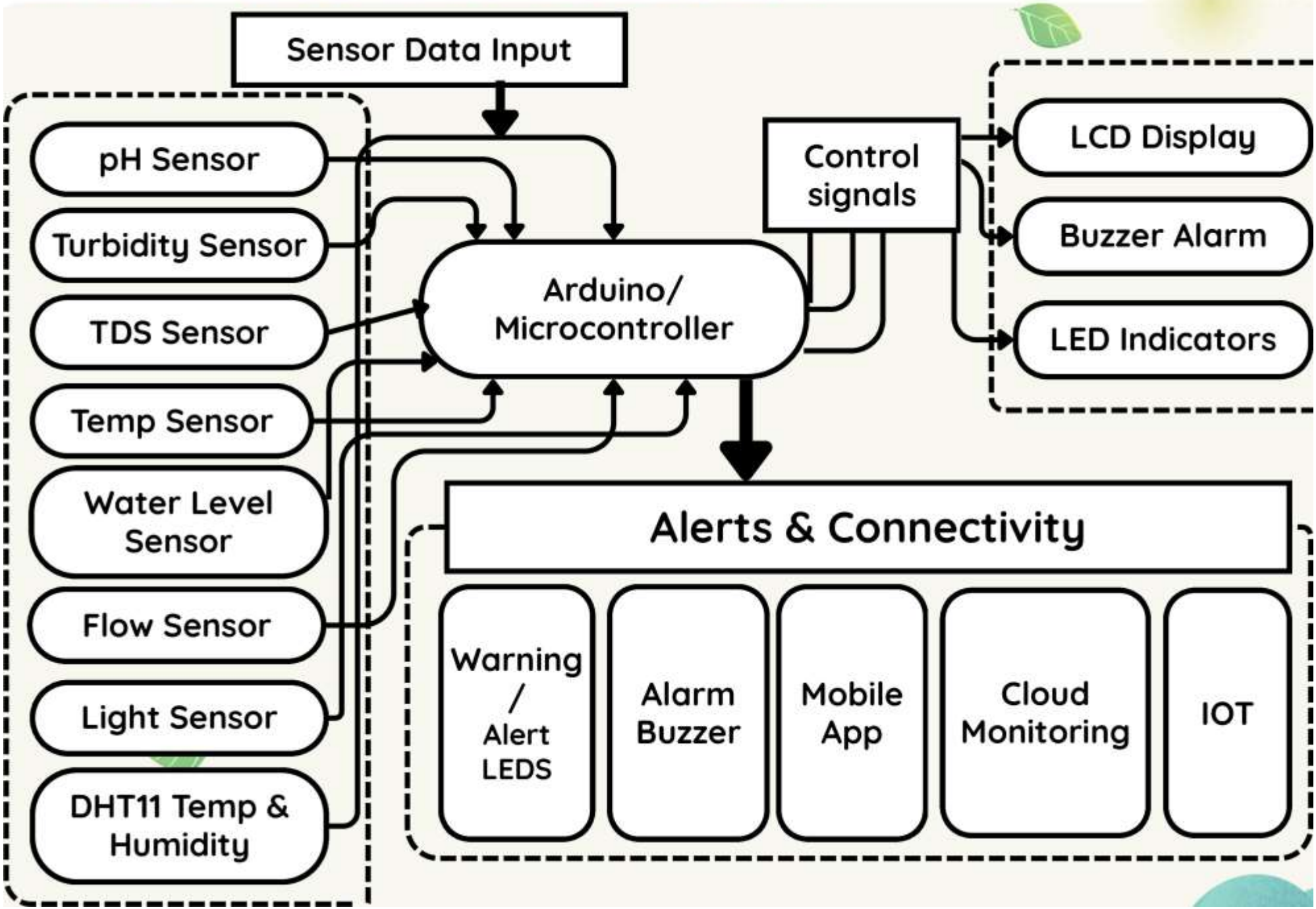
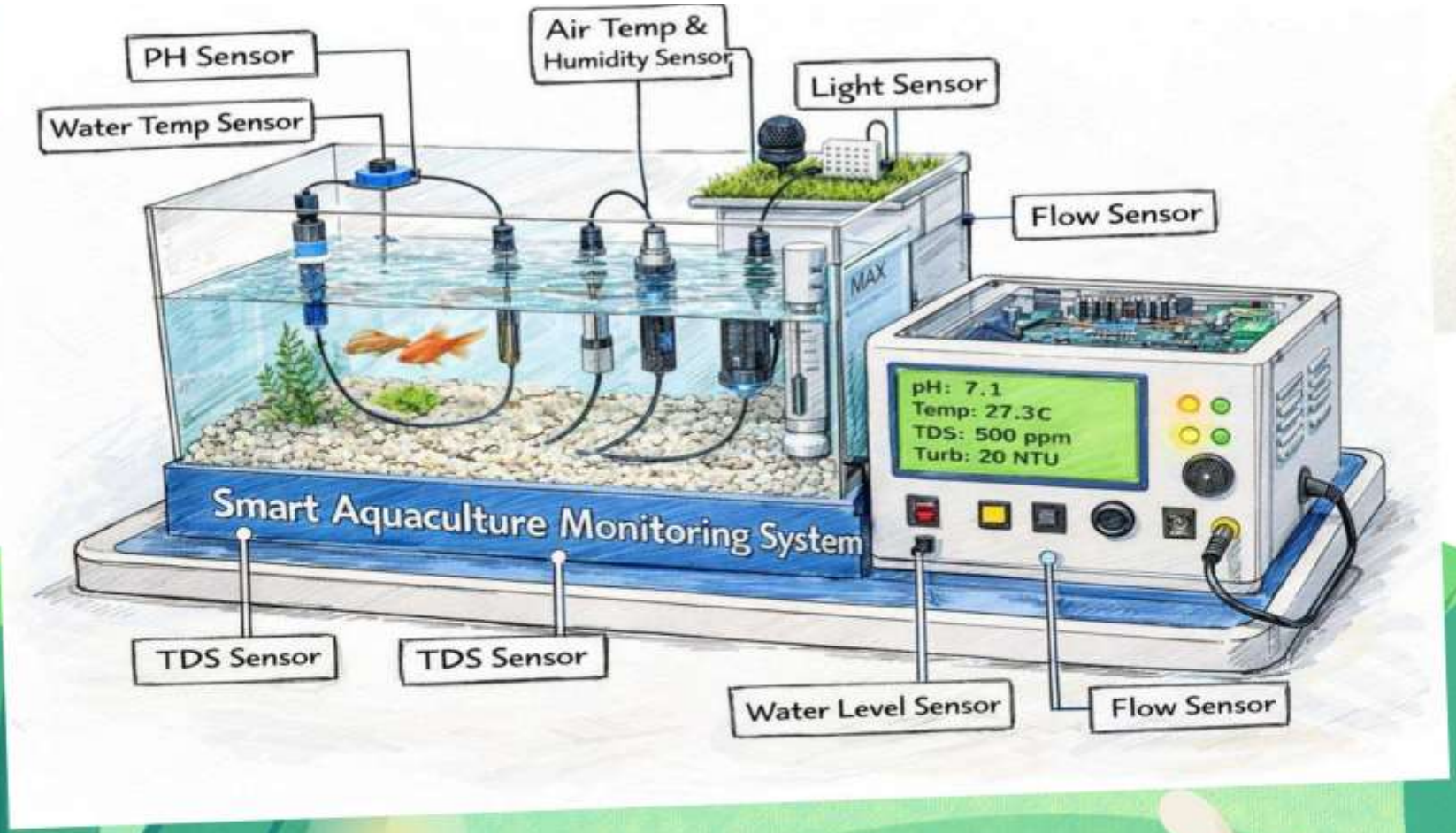
BLOCK DIAGRAM



3D MODEL


PH Sensor
Water Temp Sensor
Air Temp &
Humidity Sensor
Light Sensor
Flow Sensor
MAX
pH: 7.1
Temp: 27.3C
TDS: 500 ppm
Turb: 20 NTU
Smart Aquaculture Monitoring System
TDS Sensor
TDS Sensor
Water Level Sensor
Flow Sensor

BONPONENTS a TECHNOLOGIES

- Core: Arduino Uno/Mega microcontroller.

, Sensors:

1. pH sensor
2. TDS sensor
3. Turbidity sensor
4. Water temperature sensor
5. Water level sensor

- Outputs: LCD display, buzzer, LED indicators.
- Others: Breadboard, jumper wires, power supply, water tank.
- Software: Arduino IDE, embedded C/C++, calibration algorithms, optional IoT dashboard.

6. Flow sensor
7. Light sensor
8. Air temperature & humidity sensor (DHTII)

EXPECTED BUDGET

Component	cost (LKR)
Arduino (Uno/Mega)	2,500
pH Sensor (with probe)	4,650
TDS Sensor	3,000
Turbidity Sensor	3,500
WaterTemperature Sensor (DS18B20)	1,200
Water Level Sensor	1,500
Flow Sensor (YF-S201)	2,000
Light Sensor (LDR module)	800

Air Temp & Humidity Sensor (DHTII)			1,000
Display + OUtPUt	2,000	Tank + Materials	3,000
Miscellaneous			2,000
Total			27,150

WORK DISTRIBUTION

Member 1 - Embedded System & Power Architecture Lead

Sensors: Water Temperature + Air Temperature & Humidity Responsibilities:

- Design complete circuit architecture and power distribution (5V/3.3V regulation).
- Implement noise isolation between analog and digital circuits.

- Develop central data acquisition system and sensor communication protocols (Pc, UART).
- Synchronize multi-sensor data sampling and manage serial communication with PC.
- Timeline: Months 1-2: Study embedded systems and Arduino IDE.
  Months 3-4: Circuit and power architecture design.
  Months 5-8: Build and integrate sensor modules.
  Months 9-12: Debugging, optimization, and final integration

Member 2 - Water Chemistry & Analog Signal Engineer

Sensors: pH + TDS Responsibilities:

- Design analog signal conditioning circuits (voltage scaling, filtering).
- Develop calibration curves and compensation algorithms for sensor drift.

Implement ADC optimization and data smoothing algorithms.

Analyze water quality trends and define safe thresholds.

Timeline: Months 1-2: Research sensor characteristics.

Months 3-6: Circuit design and calibration experiments. Months 7-10: Integration and algorithm development. ' Months 11-12: Validation and fine-tuning.

Member 3 - Water Dynamics & Flow Control Engineer

Sensors: Water Level + Flow Responsibilities:

- Design pulse-based signal circuits for flow measurement.
- Develop waterproofing and insulation protection circuits.
- Implement flow rate calculation and water usage monitoring algorithms.
- Create logic for leak and overflow detection.

Timeline: Months 1-2: Sensor study and circuit design.

Months 3-6: Testing and calibration.

Months 7-10: Integration and data correction.

Months 11-12: Validation and final tuning.

Member 4 – Environmental Monitoring & Control Engineer
Sensors: Light + Turbidity Responsibilities:

- Design sensor interfacing circuits and light sensor voltage divider.
- Develop turbidity signal processing and calibration logic.
- Simulate day/night conditions using light intensity data.
- Analyze environmental impact on water clarity and fish behavior.
- Timeline: Months 1–2: Research environmental parameters.
  Months 3–6: Circuit design and sensor testing.
  Months 7–10: Integration and data interpretation.
  Months 11–12: Testing and final adjustments.

Member 5 - System Intelligence & OUtPUt Engineer

FOCUS: System logic, data fusion, and user interface Responsibilities:

- Design LCD, buzzer, and LED OUtPUt circuits.
- Develop multi-sensor data fusion and alert prioritization logic.
- Implement Al-like decision-making combining multiple parameters (e.g., pH + turbidity).
- Create user interface for real-time display and alerts.

Timeline: Months 1-2: Study OUtPUt systems.

Months 3-6: Circuit design and module development.

Months 7-10: Logic integration and UI development.

Months 11-12: System testing and demo preparation.

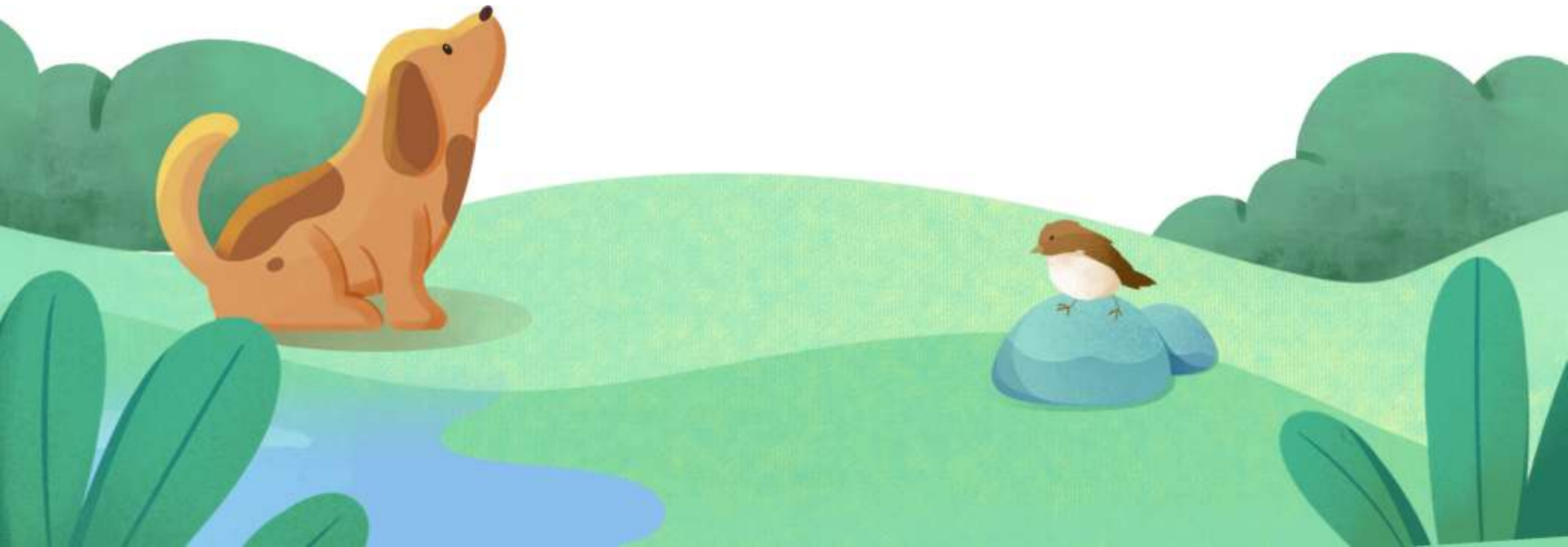
REFERENCES

Arduino Official Documentation

Research papers on aquaculture monitoring

IoT tutorials

Embedded systems notes

Sensor datasheets

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